



Master's programs' dropout and graduation clusters in a university system with a multiple enrollment policy

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ABSTRACT

Austria (Europe) allows students to enroll in multiple degree programs simultaneously or consecutively. On the master's degree level this may offer advantages for students, such as a chance for graduation instead of dropping out. This study aimed to identify groups among master's students' outcomes (graduation, dropout), considering the possibility of enrolling in multiple programs. Data from 21,412 master's degree students between 2010/11 to 2022/23 were used as a base for analyses. All possible outcome constellations of dropout and graduation have been considered and the dimensions workload and overall study duration in semesters were used for the clustering. Cluster analyses per outcome constellation, set up on the four algorithms kmeans, clara, pam and hclust, revealed differently sized clusters. Clusters were categorized into low, medium, high workload and low, medium, long duration and their frequencies were aggregated. The results suggest that enrolling in multiple programs increases overall study duration and student workload, but also that groups of students with reduced workload exist, possibly profiting from multiple enrollments. Future studies are needed to examine knowledge transfer between programs and to obtain information on differences between single-enrollment students and multiple-enrollment students in their performance in the job market.

Master's degree university students are a pre-selected group among the student population since they have already successfully finished their bachelor's degrees. Their number has increased in recent decades (Morgan, 2014). The rise in master's degrees is explained by the competitiveness of the labor market (Brooks & Everett, 2009). Financial advantages can be expected for those who graduate from master's programs (Gándara & Toutkoushian, 2017). Historical dropout rates have been reported as high as 40% in 2010 and 33% in 2023 for all OECD countries (OECD, 2010; OECD, 2023). In some other countries, a dropout rate of up to 50% was found (Arias, et al., 2023). These findings indicate that not every student will match the requirement of completing a bachelor's degree. This is supported by evidence that transitioning from bachelor's to master's programs can be socially selective (Reimer & Schwabe, 2023). Dropout rates in master's degrees are lower compared to undergraduate programs. In Germany, 7-10% are reported depending on the university type (Heublein, 2014). Other European countries show a range from 7-33% (Vossensteyn, et al., 2015). Research on master's programs has focused on different dimensions such as accessibility (Wakeling, 2017; Wakeling & Laurison, 2017) or inequalities (Posselt & Grodsky, 2003), but dropout and

retention have been and still present a research gap (Cohen & Greenberg, 2011; Rotem, et al., 2021). Austria's master's students face a unique situation since they do not have to pay tuition fees and are allowed to enroll in an unlimited number of programs simultaneously and consecutively (Federal Ministry of Education, Science and Research, 2002). This gives students the chance try out different programs and to continue studying with a chance of later graduating instead of dropping out. Although they are not directly comparable to students with multiple enrollments, graduates from double major programs have higher earnings than those with single majors, dependent on the fields of study (Del Rossi & Hersch, 2008). Therefore, it makes sense that enrolling and possibly graduating from more than one program gives students an advantage in the labor market. Evidence is needed on the outcomes of dropping out and graduating, to be able to make informed decisions on policy changes in other university systems and for future research. In Austria, no official statistics exist on this topic (Federal Ministry of Education, Science and Research, 2024). It needs to be considered that completing two or more curricula means more workload and possibly a longer time to graduation. Little is known about which outcome groups of master's students can be expected in a university system with a

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multiple enrollment policy in relation to workload and overall study time.

Austria's university policies

Austria's university system does not allocate tuition fees to its students up until the so-called "tolerance semesters" are crossed without successfully finishing a program. This timeframe is defined as two additional semesters after the minimum required time of a program. This regulation was implemented on the program level, introducing fees once one program of a student fulfills these requirements (Federal Ministry of Education, Science and Research, 2002). Given a minimum time of four semesters for most master's programs (Conference on Master-level Degrees, 2003), this means that students must pay their fees once they start the seventh semester in one of their master's degrees. Bachelor's programs can be started by attaining a general university entrance qualification (Federal Ministry of Education, Science and Research, 2002), which is usually acquired after successfully completing high school. Some of the undergraduate programs can have entrance exams that need to be passed. In general, all students are allowed to apply for admission to all undergraduate degree programs. The number of enrollments is unlimited, and they can be cross-faculty and cross-university. Starting a degree is possible prior to each semester during the official matriculation and enrollment timeframes (Federal Ministry of Education, Science and Research, 2002). Students are allowed to start multiple programs at once, add or switch programs freely as long as they do so before the semester-wise deadlines.

Master's program admission is more restricted, due to specific requirements that need to be met (Federal Ministry of Education, Science and Research, 2002). Beyond having a bachelor's degree or equivalent qualification, most master's programs require the completion of a bachelor's in the same field of study. Exceptions can be made for non-consecutive fields, similar to handling students from abroad. This entails a nostrification process to approve courses from the completed curriculum from different fields of study. Students may have to do some courses in addition to the regular master's curriculum (Federal Ministry of Education, Science and Research, 2002). Some master's programs can only be started after graduation from a consecutive bachelor's program within the same or a similar field. Other master's programs do not have any specific requirements of consecutiveness, which means that they can be started after graduation from any bachelor's program. For instance, a master's program in psychology needs a bachelor's degree in psychology as a mandatory requirement, but the master's programs Applied Ethics or Global Studies can be started with a psychology bachelor's degree or others. Both Chemistry and Molecular Biology bachelor's programs can be the base for a master's program in Biochemistry and Molecular Biomedicine. The bachelor's program in Molecular Biology enables students to start the master's degrees 1) Biotechnology, 2) Biochemistry and Molecular Biomedicine and 3) Molecular Microbiology (Federal Ministry of Education, Science and Research, 2002; University of Graz, 2021; University of Graz, 2024). Due to several different programs on the master's level in each field (University of Graz, 2021; University of Vienna, 2021), multiple enrollments are common in Austria. What possibly also facilitates and supports this behavior is the transferability of exams, meaning that exams and workload measures in ECTS credits ("European Credit Transfer and Accumulation System"), can be transferred from one program to another, once passed (Federal Ministry of Education, Science and Research, 2002). One ECTS credit is defined as 25 real-time student working hours (European Union, 2015). This can reduce curricula workload since overlapping courses or course equivalents in two curricula need only be passed once. It is not known how these regulations affect the real workload and study duration of students dropping out and graduating in different constellations.

Program type differentiation

Similar concepts of multiple enrollments exist in other university systems, which warrant differentiation. The term "dual enrollment" refers to programs that allow high school students to matriculate in the university early to do their first courses (Barnett & Stamm, 2010). These programs are known to facilitate the transition from school to university (Ison, 2022). Joint or double degree programs are implemented in two partnering institutions, which can be in different countries. Students graduate with two degrees from two institutions but are enrolled in one singular program (Dukhanov, et al., 2014; Russel, et al., 2008). Double majors are also single programs. The workload is similar to that of a regular degree program (Hemelt, 2010). Multiple enrollments in this study are defined as studying two or more curricula in a parallel or consecutive fashion. This means students must account for the full workload of each curriculum. They receive one degree for each of their successfully completed programs.

Aims and expectations

This study aims to identify groups among master's degree students' outcomes (graduation and dropout), considering the possibility of enrolling in multiple programs. Outcomes can be all possible constellations of graduation and dropout in one or more programs. Since little is known about the outcome groups in a multiple enrollment policy on the master's level, an explorative and twofold research question is defined: Using workload and study time as base dimensions, which groups of students can be identified in the unique outcome constellations? How frequent are those groups? The second aim of this study is to aid university managers' and policymakers' decisions in allowing multiple enrollments on the master's degree level. This study can be a foundation to assess the possible outcomes for students in university systems allowing for freedom of enrollment.

Methods

Data background and sample characteristics

The base data for this study was retrieved from the administrative database of the University of Graz, Austria, Europe. As Austria's second largest university, it has around 30,000 enrollments per academic year and is representative of a Mid-European university. Introduced in the early 2000s the database contains administrative student data from the campus management system of the university. Historic changes in the system, data structures, laws and other factors had an influence on the data quality. Data extraction was set to the academic years of 2010/11 to 2022/23, which was the last complete academic year in the database. This study included every study program on master's and diploma level of students with complete master's programs, i.e. from start to graduation or from start to dropout. Degree programs that did not show either of those two outcomes at the end of September 2023, i.e., those that were still enrolled, were excluded from the sample. Since the Austrian system allows for multiple programs, this study did not only consider programs on the master's level but also diploma programs. However, they were only modeled into the dataset as possible parallel programs, which means the point of view remains exclusively on master's programs. Details are explained in the next section on the data structure.

Diploma programs are studies of an old program type that was standard at Austrian universities before the Bologna reform in the 2000s, which led to the transition to the bachelor-master system. However, there are a few exceptions, e.g. law, which are still diploma programs. This type of degree program can be structured in two to three segments, has a minimum required time to graduation of eight semesters (for the case of law) and covers 300 ECTS credits (Federal Ministry of Education, Science and Research, 2002). It is a combination of bachelor's and master's programs but only awards students one degree after

graduation. Contrary to master's programs, diploma programs can be started by having acquired a formal university entrance qualification, whereas master's programs require graduation from a bachelor's degree program (Federal Ministry of Education, Science and Research, 2002). Since bachelor programs are usually defined with a minimum time to graduation of six to eight semesters and master's programs with a minimum time of four semesters (Assefa & Sedgwick, 2004; Conference on Master-level Degrees, 2003), diploma programs constitute a faster track to a college major. Students graduate with "Mag." (magister), which is the equivalent of a master's degree. Due to being on the same degree level, diploma programs were included, and the undergraduate bachelor's programs were not.

Data structure

The final dataset consisted of 21,412 students in master's programs. A three-step process was applied to the raw-data to create the final dataset for analysis. In the first step, the raw data was formatted and aggregated. Each row represented one master's program of a given student. The ID column for the programs was defined as the "observed program". If a student had two or n more master's programs during their time at the University of Graz, they had two to n rows. In the second step, parallel and consecutive programs were identified and merged with the initial data. They were defined either as programs with overlapping timeframes or consecutive programs. The overlap means that the enrollment status of two programs overlapped at least in one semester. Consecutiveness was defined as the situation of starting a second program after the end of the first one, with the limitation of enrolling no more than one academic year apart. This can be the case if students switch their master's program, i.e., enroll in a new one, and close their previous one mid-semester. Due to official matriculation timeframes, the start of a second program may not be possible before the next year's winter semester. All these cases were treated the same way. Programs that had more time between their start and end semesters were not defined as parallel programs in the dataset. For those students with more than one program, the data matching process generated a second column "multiples", where parallel and consecutive programs were shown. In this step, the merging function acted in a bidirectional manner, which means that two parallel master's programs of the same student were shown as two separate rows for each "observed program", each with the other program in the "multiples" column. This included both master's and diploma programs. In the final step, rows with diploma programs as the observed programs were removed to keep the focus on the master's programs. Additionally, the observed programs were set to a neutral value and a distinct function was applied to the dataset. This method minimized the impact of the multiplication effect as all programs with one additional program were reduced to one row. However, this could not account for constellations of more than two parallel or consecutive programs, which were still represented as multiple rows. This was necessary, as different constellations meeting the inclusion criteria could lead to different constellations of parallel programs. For instance, a student with three master's programs could have started program 1 at the same time as program 2. Program 2 can be closed after one semester and a third program can be started during the fourth semester of program 1. This makes program 1 both parallel to 2 and 3, but there is no parallel connection between the latter two. In the raw dataset, there would be four rows for parallel constellations 1×2 , 1×3 , 2×1 and 3×1 . In the final dataset, rows for 1×2 and 1×3 remain. Descriptive analyses of the final dataset showed that 18,759 students had no parallel program (73%), 4,950 rows (19%) were single rows of students with two parallel programs, 1,545 students (6%) had at least one extra row for a third program and that 519 students (2%) had more than two parallel program constellations. When interpreting the results of this study, this program level structure needs to be kept in mind.

Variables

All analyses were done dependent on the outcome. A variable for the outcome status in each row in the dataset was generated, i.e., showing "graduation" and "dropout" for single-program rows or the parallel constellations "dropout-dropout", "graduation-dropout" and "graduation-graduation" for rows of students with more than one program.

The major variables for cluster analyses were ECTS credits and the overall duration of the degree programs in semesters. Both measures were summarized across each program constellation per row, for students with more than one enrollment. The duration variable was calculated, considering the overlap between two programs as the same study time. This means the duration in semesters is measured from the beginning of the program that was started first to the end of the program that was ended last. Overlapping semesters were counted as 1. A student with two programs should be used as an example: program 1 ended in the third semester. However, program 2 was started during the second semester of program 1. Then, program 2 ended one semester after program 1 was closed. Thus, four semesters are counted. The duration represents the overall time a student spent in the university system enrolled in a program constellation, i.e., one row in the dataset. ECTS credits must have originated from real exams to be counted. If credits are transferred from one program to another as elective or mandatory course substitutes, ECTS credits were only counted in the program the exam was originally passed in. Also, credits originating from the master's thesis were not included. This is why the mean total ECTS credits can have smaller values than the minimum credits to finish a curriculum. The cluster analysis is based on these two dimensions for workload and time until an outcome status is reached. Table 1 shows descriptive statistics for these variables per outcome status.

Apparatus

Data curation and analyses were executed on a PC with a Windows 11® installation, 16 Core, 64 GB RAM, using R (R Core Team, 2022). Raw data was queried from an Oracle® SQL database using the Oracle InstantClient 19® and the R-package RODBC (Ripley & Lapsley, 2022). Cluster analyses and base visualizations were calculated via the cluster and factoextra packages (Kassambara & Mundt, 2020; Maechler, et al., 2023). Further processing and customization of the visualizations was done via the ggplot2 package (Wickham, et al., 2023).

Statistical Analyses, Algorithms and Settings

Cluster analysis was chosen as preferred method to group students' outcomes (Ammon, et al., 2008). The analyses were done separately for each outcome, which means that subsets of the data were generated based on the outcome level. This was handled automatically, by the code that was designed to loop through the outcome levels and algorithms. Subsequently, the four algorithms "clara", "hclust", "kmeans" and "pam" were applied to each of the datasets resulting in six outcome levels x four analyses each. At first, the code filtered the outcome level and determined the algorithm to be used. Both dimensions, ECTS credits and duration, were standardized. Then, the average silhouette method was executed to find the optimal number of clusters. At last, the analysis

Table 1

Descriptive statistics of the cluster dimensions ECTS credits and duration in semesters.

Outcome	<i>M</i> ECTS	<i>SD</i> ECTS	<i>M</i> duration	<i>SD</i> duration
Dropout	18.69	28.31	10.64	6.73
dropout-dropout	16.11	25.67	8.12	5.28
Graduation	80.67	25.66	9.83	5.46
graduation-dropout	48.86	39.5	9.58	4.18
graduation-graduation	78.33	22.65	11.9	3.34

with the specific algorithm was run and plots were generated.

Determining the optimal number of clusters via the average silhouette method a matrix of $n \times 3$ was generated in the process, with n as the number of rows per dataset (Maechler, et al., 2023), which led to initial memory issues of the hardware during programming. To bypass these limitations, the code was configured to choose a random sample of 1,000 rows from each subset larger than this. The visualizations for the average silhouette per subset can be found in the supplementary material. The use of this adjustment is indicated by an annotation in the bottom left corner of the plots. In general, the average silhouette method determines how well individual objects, i.e., data points, lie within each cluster. Additionally, it examines which objects lie merely between clusters. The results are then used to find an appropriate number of clusters (Rousseeuw, 1987).

The following algorithms have been used in this study:

- kmeans: this algorithm minimizes the sum of squared Euclidean distances from objects to the assigned cluster centers and partitions the objects into k groups. The Hartigan-Wong algorithm is used, `nstart` is set to 25 and maximum iterations to 1,000 (R Core Team, 2022; Hartigan & Wong, 1979).
- pam – partitioning around medoids: this is a more robust version of kmeans, based on minimizing the sum of dissimilarities. So-called medoids represent clusters, with the algorithm searching for k representative ones among the objects of the dataset. After defining medoids, the datapoints are assigned to their nearest medoid, by minimizing the sum of dissimilarities to the closest medoids (Kaufman & Rousseeuw, 2009; Maechler, et al., 2023). In the function settings, metric is set to “euclidean”.
- clara – clustering large applications: this is the same algorithm as pam, but was specifically designed to deal with larger datasets. It relies on subsets of the input data of fixed sizes, which reduces memory load. Cluster quality is assessed by the mean of dissimilarities of the observation to the closest medoid, retaining the minimum values (Kaufman & Rousseeuw, 2009; Maechler, et al., 2023). The metric was set to “euclidean” and the number of samples to 25.
- hclust: this hierarchical clustering method is contained in the `hcut` package, i.e., “computes hierarchical clustering and cut the tree” (Kassambara & Mundt, 2020). The algorithm is iterative and repeatedly assigns each object in the dataset to a cluster, computes distances between clusters in the iterations and gets the results from combining the most similar cluster at each level until one is left (R Core Team, 2022). The metric “euclidean” was used and the method was set to “average”.

The results of the analyses were aggregated per outcome in ECTS and duration groups. A 3×3 matrix of ECTS clusters $\times$ duration clusters was set up and the outcomes were counted per group. ECTS groups low, medium and high and duration groups of low, medium and long were created. They were based on the idea that master’s programs cover 120 ECTS credits and have a minimum time of four semesters (Conference on Master-level Degrees, 2003). A package of first-semester courses is equivalent to 30 ECTS credits, which was used as the mark for the low ECTS group. Medium was defined in between, from >30 to <60 ECTS credits. Having completed more than two semesters means around 60+ ECTS credits, with the upper boundary being therefore defined as 60+ ECTS. The group labeled “low duration” was defined as students that did not stay in the system up to the minimum required time of four semesters. Medium was defined as four to six semesters, which is the tolerance timeframe in Austria, before students must pay fees (Federal Ministry of Education, Science and Research, 2002). Everything beyond six semesters was considered long. The final results were obtained by counting the total n and transforming them into relative frequencies.

Results

Cluster analyses

Cluster analyses were executed separately for each outcome and algorithm, i.e., 24 distinct calculations. Descriptive statistics of the results for each cluster can be found in Table 2. Model parameters for each algorithm and outcome are listed in Tables A1-A4 in the appendix. Average silhouette plots and cluster plots are depicted in the supplementary material.

Clusters for single enrollment students with the outcome dropout showed two clusters in all algorithms. The `hclust` algorithm identified a third cluster with two cases. The outcome graduation from one program also shows three clusters in every algorithm. However, one of these clusters only had one case.

Students with more than one enrollment and the outcome dropout-dropout were clustered into two or three clusters. `Hclust` created a cluster with a low number of cases also for this outcome. Graduation-dropout showed three clusters, with the exception of `hclust` creating five. The same pattern was found for graduation-graduation which was categorized into two clusters, except for `hclust` partitioning the data into five clusters with one single-case result.

Following the model parameters of the kmeans algorithm (Table A1), the two dimensions ECTS credits and duration could explain variance near and above 50% for almost all models. Dropout-dropout showed 45% and graduation-graduation 33%. The maximum could be obtained for graduation-dropout with 65%.

Cluster aggregation

The total n was summed up over all algorithms per individual outcome and the results are shown in Fig. 1, listing both absolute and relative frequencies. Dropout clusters were most identified as low ECTS-low duration. The clusters in the low ECTS-low duration and medium ECTS-long duration groups were less frequent. Dropout-dropout clusters were most frequently identified as low ECTS-long duration or low ECTS-medium duration. In the graduation clusters, the most frequent group was high ECTS-long duration, followed by medium ECTS-medium duration and medium ECTS-long duration. Graduation-dropout clusters only had results in the long duration column, with varying frequencies for ECTS. Graduation-graduation clusters showed a very large group of high ECTS-long duration and one small group of low ECTS-long duration.

Discussion

Outcome clusters

This study aimed to identify groups among master’s degree program students’ outcomes (graduation and dropout), considering the possibility of enrolling in parallel and consecutive programs. The first part of the research question was about which groups of students can be identified in the unique outcome constellations dependent on their workload and study duration. The second part addressed the frequency of the groups found.

Drop out

Both consistent and inconsistent numbers of clusters have been identified by the algorithms for each outcome. The `hclust` algorithm in the `hcut` package showed more inconsistency than others, creating small and single-person clusters in addition to its larger ones. The other algorithms found similar and reasonably sized clusters. To overcome this algorithm behavior, results were aggregated, and relative frequencies of clusters were calculated. The results show that dropout clusters of students in one program have the highest frequency in the low ECTS-low duration group. Most students drop out fast with a minimal workload,

Table 2

Cluster and descriptive statistics based on the dimensions ECTS and duration in semesters.

Outcome	cluster	method	ECTS	SD ECTS	duration	SD duration	n
Dropout	1	clara	6.14	9.47	3.86	2.43	3.508
	2		58.27	31.33	11.65	5.62	1.113
	1	hclust	12.60	19.70	4.95	4.00	4.218
	2		81.86	24.37	13.96	4.99	401
	3		206.75	10.25	15.50	3.54	2
	1	kmeans	6.39	9.90	3.83	2.36	3.519
	2		57.98	31.94	11.80	5.58	1.102
	1	pam	5.54	8.55	3.74	2.30	3.410
	2		55.73	31.52	11.35	5.57	1.211
dropout-dropout	1	clara	5.71	8.95	4.79	2.36	2.877
	2		6.68	8.75	14.12	3.67	1.108
	3		66.06	22.70	11.78	4.98	808
	1	hclust	15.86	25.00	8.12	5.28	4.784
	2		150.83	24.42	12.00	1.94	9
	1	kmeans	45.45	33.12	13.93	5.05	1.262
	2		5.62	9.20	6.05	3.51	3.531
	1	pam	5.25	8.46	4.67	2.29	2.788
	2		8.19	10.13	13.86	3.77	1.222
graduation	3		67.15	22.23	11.47	4.96	783
	1	clara	59.31	19.09	5.87	1.78	1.535
	2		98.92	15.13	6.91	1.73	1.785
	3		81.18	21.90	15.34	3.68	423
	1	hclust	80.50	25.75	6.69	2.17	3.478
	2		82.45	23.26	17.26	3.33	264
	3		208.50		8.00		1
	1	kmeans	83.23	22.35	15.86	3.56	377
	2		96.36	15.52	6.81	1.77	2.049
graduation-dropout	3		55.53	18.39	6.00	1.98	1.317
	1	pam	56.54	18.68	5.91	1.81	1.352
	2		97.35	15.47	6.45	1.42	1.801
	3		85.06	21.13	13.94	3.84	590
	1	clara	49.61	35.15	15.04	2.91	2.854
	2		9.45	12.10	7.60	2.33	3.565
	3		84.22	20.47	7.42	2.24	3.914
	1	hclust	85.48	16.97	14.14	1.83	1.104
	2		12.86	15.55	9.41	3.77	4.971
graduation-graduation	3		83.07	19.58	7.44	2.21	3.876
	4		57.43	29.64	20.58	1.68	354
	5		154.21	15.61	18.29	2.57	28
	1	kmeans	10.06	12.83	7.91	2.55	3.863
	2		50.78	34.58	15.61	2.82	2.379
	3		84.39	20.07	7.66	2.41	4.091
	1	pam	59.41	33.25	14.92	3.10	2.762
	2		8.98	11.61	8.17	2.74	3.923
	3		83.78	20.54	7.07	2.01	3.648
	1	clara	89.07	16.87	13.32	2.89	4.480
	2		60.87	19.79	9.60	2.69	2.756
	1	hclust	81.68	18.62	11.85	3.03	6.663
	2		27.70	12.42	9.27	2.84	411
	3		64.21	25.69	20.83	2.20	156
	4		189.40	13.84	11.60	1.67	5
	5		196.00		20.00		1
	1	kmeans	61.48	19.76	9.67	2.68	2.872
	2		89.41	16.84	13.37	2.89	4.364
	1	pam	90.03	16.80	13.45	2.90	4.176
	2		62.35	19.63	9.79	2.68	3.060

similar to early dropouts in bachelor's programs (OECD, 2019). Higher frequencies can be found in groups with low and medium ECTS-medium duration. No groups were identified for low ECTS-long duration. This study cannot give definitive answers on the reasons for dropping out, but previous research gives suggestions, on which groups of students may be contained within the clusters identified in this study. The results show that students either do not accomplish any workload and leave early or they try to accomplish workload but leave before they must pay tuition fees in the seventh semester (Federal Ministry of Education, Science and Research, 2002). This indicates that students try to avoid financial pressure, which is known to be a reason for dropping or stopping out (Terriquez & Gurantz, 2015). Previous research pointed to financial aid as a key predictor for retention in master's programs (Rotem, et al., 2021), originating from work on doctoral studies (Ampaw & Jaeger,

2012). A previous study on two master's programs and distance learning showed that age can be a predictor for dropout. Students are more likely to do drop out with increasing age (Patterson & McFadden, 2009). By being older than undergraduate students and not having the same opportunities for funding as doctoral students, master's degree students' may leave the university due to work and family obligations or other dimensions correlated with increasing age (Rotem, et al., 2021). Other research on master's programs categorized influencing dimensions of student dropout. Work, family and personal obligations, institutional difficulties, the program's economic feasibility and harassment were identified (Rotem, et al., 2021). However, the same study found that academic variables predict dropping out better than the other variables (Rotem, et al., 2021). Course failure is a general predictor for leaving university (Nora & Crisp, 2012), which may explain the medium ECTS

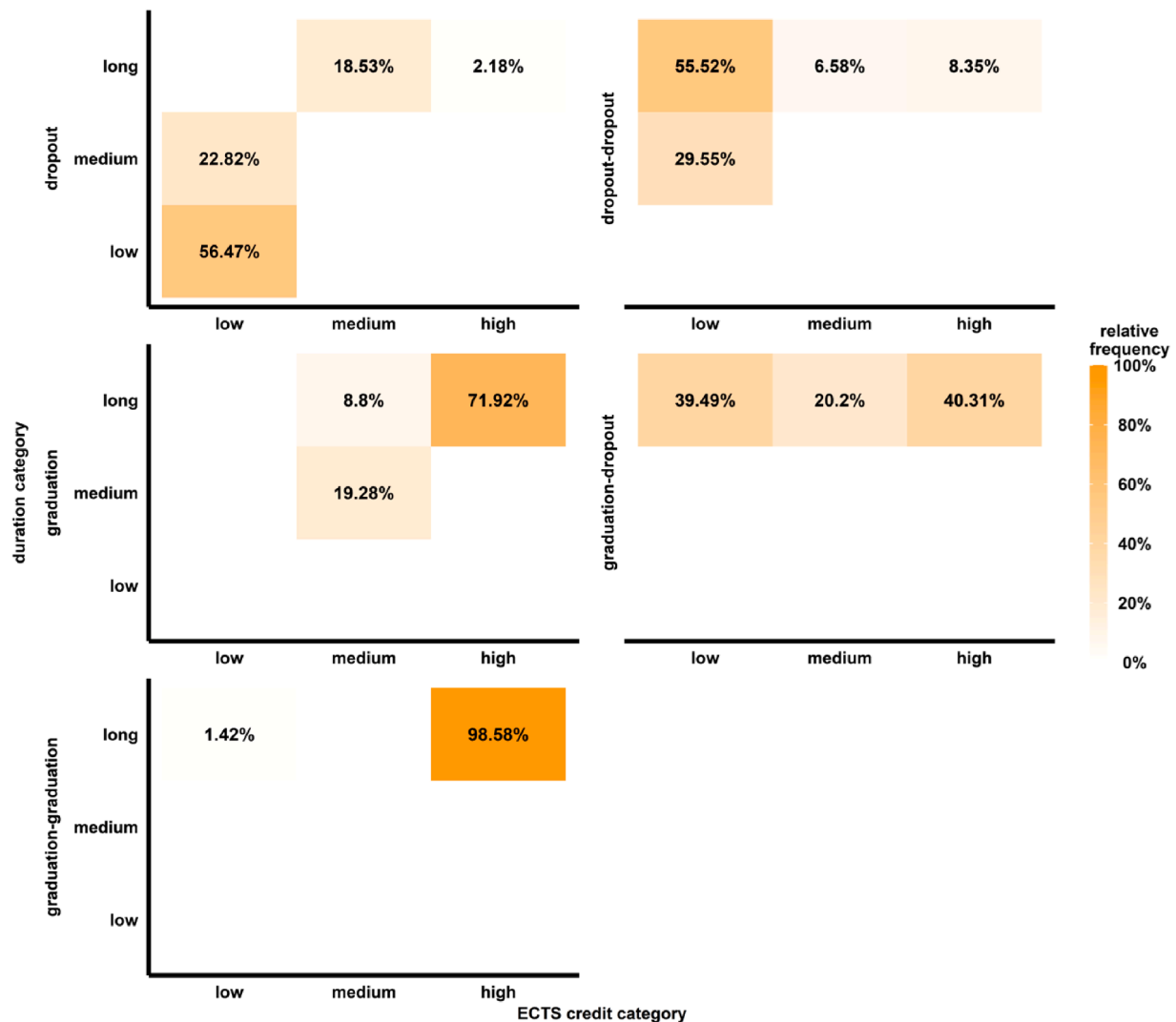


Fig. 1. Cluster aggregation in absolute and relative frequencies of the total n over all algorithms per outcome.

groups. Performance predictions of master's students are more accurate after the second semester than earlier (Alturki, et al., 2022), which may be due to the early dropout identified in this study.

Program constellations with dropout-dropout as outcome had no groups in the low duration segments. Medium and long duration were the most common, both in low ECTS groups. Medium and high ECTS groups could also be identified, but were less common. Compared to single program dropout, this shows that students have a longer overall study duration when doing more than one program. This study cannot dive deeper into the characteristics of these students, as psychological and behavior-related variables are missing in the database. Based on previous research that not every student may have the affective and cognitive abilities to graduate from university (Díaz Mújica, et al., 2019), this can be one group among dropout-dropout clusters. They may show a stronger will to persist than those starting only one program and be more likely try out different ones, although leaving in the end. Persistence is a dimension related to master's degree students' retention, with personal factors, e.g., support from family, being more important than institutional ones, e.g., university services (Cohen & Greenberg, 2011). Social and academic integration play a role in master's students' persistence (Hammond & Shoemaker, 2014). This may help some students with similar characteristics as the group of dropout-dropout students to graduate, ultimately resulting in the graduation-dropout groups. Another group may attend university, possibly without the wish

to graduate, looking for micro-credentials (Selvaratnam & Sankey, 2021) by trying out different curricula and only doing courses to improve their skills. Identifying the reasons for multiple enrollments regarding different outcomes needs to be the focus of future research.

Graduation

Graduation groups showed high frequencies in the high ECTS-long duration segment. Medium ECTS-medium duration and medium ECTS-long duration were less frequent. No other clusters were identified. Given 120 ECTS credits to finish a master's degree (Conference on Master-level Degrees, 2003), medium groups can be explained by not including ECTS credits from master theses. The theses in different curricula at the University of Graz are worth 20+ ECTS credits, e.g., Psychology with 28 ECTS credits, Biotechnology with 30 ECTS credits or Business Economics with 20 ECTS credits (University of Graz, 2024). Therefore, completing a master's program without doing additional courses outside of the curriculum can mean falling either in the medium or high ECTS categories.

Both clusters for the outcomes graduation-dropout and graduation-graduation were identified as groups with long duration. While graduation-graduation had almost exclusively assignments to the high ECTS-long duration group, graduation-dropout groups showed distinct frequency values over all three ECTS categories. The clusters found in the low ECTS segment can be explained by 1) the program constellations

not incorporated in the analyses and 2) bachelor's students taking master's courses. 1) Due to each row in the dataset having a maximum program constellation of two programs, ECTS credits from a third "main" program could not be considered in the analyses. Workload was only counted once for the program the exams were done in. Cross-program transfer and transfer of courses from semesters abroad could not be accounted for. There was also no possibility to model cross-university credit transfer for Austrian universities since the databases are not connected to each other. 2) In most undergraduate programs, it is possible to take exams and attend courses at the master's level. As long as there are empty spaces, bachelor's students can take master's courses early, for instance during the last few semesters before graduation, when they are already working on their theses and no other bachelor's exams are left to be done. The data structure of this study also did not account for individual exams, which means that an exact matching of students' curricula and passed exams across all their past enrollments would have been necessary. Therefore, the analyses show the real workload students had to accomplish per program constellation in those programs.

The results indicate that possible transfer can reduce the workload in a curriculum. Undergraduate performance being able to predict students' performance at the master's level supports these findings (Zimmermann, et al., 2015). The more successful students are in their bachelor's programs, the easier it could be for them to attend master's courses early, which affects the workload they have left to accomplish during their master's programs. It needs to be considered that relatedness of the undergraduate and graduate programs is important for students' outcomes. A high relatedness between programs shows relationships between bachelor's and master's grades twice as strong compared to a low relatedness (Peters & Daly, 2013). This makes the choice of undergraduate programs as well as which (multiple) master's programs are chosen important for success. For multiple enrollments, this means that those with a higher relatedness between their master's programs may have higher chances of graduating from one or more programs compared to others. One study on online psychology master's programs in the US found that prior graduate school experience is associated with a lower probability of graduation (Newhouse & Cerniak, 2016). This may indicate that graduation may also be harder for those students adding a second program in their later semesters. Since the analyses allow no causal relationship to be established (Newhouse & Cerniak, 2016), this can mean that one reason for this effect can be students at risk of dropping out switching their programs and trying to be successful in another program. Another reason could be the increased workload of a second program or obligations that come with increasing age (Rotem, et al., 2021).

Methodological considerations and limitations

Since this study was set up on the program level, i.e., a maximum of two programs per row, not on the student level, ECTS transfer from one program to another, which was not present in a given row, could not be accounted for. Also, program sequences more than one academic year apart were not considered consecutive and modeled in the dataset. However, transfer of ECTS credits and exams from these programs is also possible (Federal Ministry of Education, Science and Research, 2002). Such constellations not being the norm for students, the impact on the results may be low.

The variance that could be explained by the dimensions ECTS credits and study duration was moderate to high for most groups, except for graduation-graduation. This suggests that there are other outside factors that may explain the clusters better. Similar to the choice of one's field of study being a multidimensional construct (Alonderiene & Klimavičiene, 2013), the reasons for individual outcomes of students in more than one program may also be multifactorial. Two dimensions may not be enough to explain the variance of each cluster.

The hclust algorithm in the hcut package was not found to be ideal for clustering student outcomes. Single-person and small clusters were

created, with other algorithms identifying larger-sized clusters. In terms of clustering students using the dimensions of workload and study time, other algorithms should be considered.

Future outlook and implications

With respect to the identified clusters, it needs to be considered that those who drop out from one or more programs may profit from a knowledge transfer that facilitates graduation in a second program or increases their chances in the job market. Previous work shows that prior knowledge is an indicator of academic performance (Seery, 2009). Declarative knowledge about a field of study may not always be transferable by students, e.g., from Global Studies to Biotechnology, but students may make use of the procedural knowledge they acquired in a parallel or previous master's program. This type of knowledge is also associated with performance (Hailikari, et al., 2007; 2008). This can also be the case for metacognition, which is involved in planning and creating strategies for studying (Abdelrahman, 2020). However, one study indicates that it may be harder to graduate from a program if there has been a prior enrollment on the same degree level (Newhouse & Cerniak, 2016). It also needs to be taken into consideration that students who attended college have higher chances in the job market compared to their peers, who have never entered the university system (Giani, et al., 2020). Since some employers hire people based on job-relevant skills more so than completed degrees (Neugebauer & Daniel, 2022), a bachelor's degree and specific courses from one or more master's programs could be enough to have an advantage over other bachelor's graduates. However, both knowledge transfer and job market performance of students in one or more programs need closer consideration in future studies.

The results of this study have implications for policymakers and university management about multiple enrollment policies of master's programs. Overall study duration and workload are higher for students in multiple enrollments compared to single enrollments. The results also suggest that there can be an effect of reduced workload in a second program if prior work is transferrable (Federal Ministry of Education, Science and Research, 2002). Previous research indicates that multiple programs should have a high relatedness to each other as well as to the undergraduate program of students, for increased success (Peters & Daly, 2013). Considering students with one program leave earlier than those with multiple enrollments independent of the outcome, the Austrian university system may need targeted support structures for those students (Linden, 2022), which are currently not present. In other regions, the decision to allow multiple enrollments should be based on the individual university system. Austria does not allocate fees to students (Federal Ministry of Education, Science and Research, 2002), which may be a reason to experience financial pressure for students in other systems. Due to the results of this study indicating that multiple programs are associated with longer study duration, students may drop out or not consider paying for another program after several semesters of unsuccessful studying (Terriquez & Gurantz, 2015). The transferability of exams that Austria offers (Federal Ministry of Education, Science and Research, 2002) may also be a critical policy to be considered in other systems.

Conclusions

This study identified different groups of outcomes among students in master's programs, being allowed to enroll in multiple programs in a parallel and consecutive fashion. Future studies are needed to examine a possible knowledge transfer from one program to another and to obtain information on differences between single-enrollment students and multiple-enrollment students in their performance in the job market. The findings of this study suggest that enrolling in multiple programs increases overall study duration and student workload. However, there also exist groups of students with reduced workload because of transfer

from one program to another. Targeted support structures for students with multiple enrollments may be needed. Transferability of exams and no additional fees may be considered by policymakers on introducing a multiple enrollment policy.

Ethics approval

the analysis of the data and the permission for publication have been approved in a works agreement between the Performance- and Quality Management Department and the works committee of the University of Graz, called “Rahmen-BV IKT 2019”, which includes ethics approval. Furthermore, data analyses on student data by staff departments of the university is regulated by law, defining analyses as a means of increasing graduation rates of the Austrian universities as an obligation ([Federal Ministry of Education, Science and Research, 2002](#); see references in manuscript).

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Supplementary materials

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Appendix A. Model parameters

Table A1
Model parameters for the kmeans algorithm.

outcome	cluster	Variable	value	n	total SS	total within SS	within SS	between SS	iterations	variance explained
dropout	1	total ECTS credits (standardized)	-0.43							
	2		1.39							
	1	total time in the university system	-0.39							
	2	(standardized)	1.26							
	1	model parameters		3,519	9,240	4,156.21	1,277.69	5,083.79	1	55.02%
	2			1,102			2,878.52			
dropout-dropout	1	total ECTS credits (standardized)	1.14							
	2		-0.41							
	1	total time in the university system	1.1							
	2	(standardized)	-0.39							
	1	model parameters		1,262	9,584	5,272.5	3,254.52	4,311.5	1	44.99%
	2			3,531			2,017.98			
graduation	1	total ECTS credits (standardized)	0.1							
	2		0.61							
	3		-0.98							
	1	total time in the university system	2.39							
	2	(standardized)	-0.18							
	3		-0.41							
	1	model parameters		377	7,484	3,020.45	667.63	4,463.55	3	59.64%
	2			2,049			1,264.04			
	3			1,317			1,088.78			
graduation-dropout	1	total ECTS credits (standardized)	-0.98							
	2		0.05							
	3		0.9							
	1	total time in the university system	-0.4							
	2	(standardized)	1.44							
	3		-0.46							
	1	model parameters		3,863	20,664	7,173.51	1,846.43	13,490.49	3	65.29%
	2			2,379			2,907.88			
	3			4,091			2,419.19			
graduation-graduation	1	total ECTS credits (standardized)	-0.74							
	2		0.49							
	1	total time in the university system	-0.67							
	2	(standardized)	0.44							
	1	model parameters		2,872	1,4470	9,708.37	4,036.9	4,761.63	1	32.91%
	2			4,364			5,671.47			

Graz.

Data availability

Data cannot be made available due to being a large size dataset of internal university administrative data.

CRediT authorship contribution statement

Alexander Karl Ferdinand Loder: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

the author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Table A2

Model parameters of the clara algorithm.

outcome	cluster	variable	value	<i>n</i>	<i>Max</i> diss	<i>M</i> diss	isolation	<i>Med</i> iteration	cluster <i>M</i> width	<i>M</i> width	objective
dropout	1	total ECTS credits (standardized)	-0.48								
	2		1.57								
	1	total time in the university system	-0.57								
	2	(standardized)	1.09								
	1	model parameters		3,508	2.29	0.49	0.87	4,406	0.72	0.65	0.72
dropout-dropout	2			1,113	5.53	1.42	2.1	2,385	0.4		
	1	total ECTS credits (standardized)	-0.63								
	2		-0.63								
	3		1.79								
	1	total time in the university system	-0.59								
	2	(standardized)	0.92								
	3		0.92								
	1	model parameters		2,877	1.93	0.51	1.27	1007	0.52	0.53	0.66
	2			1,108	2.35	0.69	1.55	3,556	0.53		
	3			808	5.93	1.17	2.46	3,193	0.53		
graduation	1	total ECTS credits (standardized)	-0.73								
	2		0.71								
	3		-0.08								
	1	total time in the university system	-0.41								
	2	(standardized)	-0.12								
	3		2.14								
	1	model parameters		1,535	2.8	0.76	1.91	3,221	0.37	0.45	0.76
	2			1,785	4.28	0.66	2.91	3,325	0.48		
	3			423	3.57	1.19	1.49	2,966	0.52		
graduation-dropout	1	total ECTS credits (standardized)	0.08								
	2		-1.11								
	3		0.89								
	1	total time in the university system	1.3								
	2	(standardized)	-0.62								
	3		-0.62								
	1	model parameters		2,854	3.13	1.04	1.5	1,767	0.27	0.51	0.74
	2			3,565	1.68	0.58	0.84	3,241	0.62		
	3			3,914	3.5	0.67	1.75	2,233	0.59		
graduation-graduation	1	total ECTS credits (standardized)	0.32								
	2		-0.68								
	1	total time in the university system	0.33								
	2	(standardized)	-0.57								
	1	model parameters		4,480	5.56	0.94	4.16	6,581	0.27	0.37	0.97
	2			2,756	3.89	1.01	2.91	5,331	0.56		

Table A3

Model parameters of the pam algorithm.

outcome	cluster	variable	value	<i>n</i>	<i>Max</i> diss	<i>M</i> diss	diameter	Separation	<i>Med</i> iteration	cluster <i>M</i> width	<i>M</i> width	objective build	objective swap
dropout	1	total ECTS credits	-0.55										
	2	(standardized)	1.35										
	1	total time in the	-0.57										
	2	university system	0.89										
	1	(standardized)											
dropout-dropout	1	model parameters		3,410	2.08	0.46	2.89	0.02	225	0.75	0.61	0.74	0.71
	2			1,211	5.79	1.41	7.7	0.02	2,401	0.22			
	1	total ECTS credits	-0.51										
	2	(standardized)	-0.39										
	3		1.87										
	1	total time in the	-0.59										
	2	university system	0.92										
	3	(standardized)	0.54										
	1	model parameters		2,788	1.77	0.48	2.29	0.02	553	0.61	0.51	0.72	0.64
	2			1,222	2.33	0.7	3.09	0.02	118	0.41			
graduation	3			783	5.89	1.13	7.14	0.02	3,367	0.33			
	1	total ECTS credits	-0.73										
	2	(standardized)	0.54										
	3		0.21										
	1	total time in the	-0.41										
	2	university system	-0.41										
	3	(standardized)	1.58										
	1	model parameters		1,352	2.8	0.75	3.79	0.02	1,200	0.34	0.39	0.81	0.75
	2			1,801	4.48	0.61	5.19	0.02	287	0.47			

(continued on next page)

Table A3 (continued)

outcome	cluster	variable	value	n	Max diss	M diss	diameter	Separation	Med iteration	cluster M width	M width	objective build	objective swap
graduation- dropout	3			590	3.96	1.18	6.13	0.06	234	0.22			
	1	total ECTS credits	0.36										
	2	(standardized)	-1.06										
	3		0.85										
	1	total time in the	1.06										
	2	university system	-0.38										
graduation- graduation	3	(standardized)	-0.62										
	1	model parameters		2,762	3.46	1.01	5.59	0.01	2,533	0.23	0.44	0.87	0.73
	2			3,923	1.92	0.64	3.35	0.01	62	0.53			
	3			3,648	2.37	0.63	3.89	0.01	3,682	0.52			
	1	total ECTS credits	0.45										
	2	(standardized)	-0.54										
	1	total time in the	0.33										
	2	university system	-0.57										
		(standardized)											
	1	model parameters		4,176	5.43	0.94	8.7	0.02	1,284	0.34	0.31	1.01	0.97
	2			3,060	3.96	1.01	7.17	0.02	134	0.28			

Table A4
Model parameters of the hclust algorithm.

outcome	cluster	n	cluster M width	M width
master's programs dropout	1	4,218	0.61	0.59
	2	401	0.47	
	3	2	0.75	
dropout-dropout	1	4,784	0.66	0.66
	2	9	0.82	
graduation	1	3,478	0.55	0.54
	2	264	0.49	
	3	1	0	
graduation-dropout	1	1,104	0.45	0.38
	2	4,971	0.34	
	3	3,876	0.40	
	4	354	0.41	
	5	28	0.59	
graduation-graduation	1	6,663	0.35	0.36
	2	411	0.54	
	3	156	0.47	
	4	5	0.59	
	5	1	0	

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